

Breaking the Classics Laboratory Report

A Cryptography Escape Room

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Introduction

This lab is an escape-room-style exercise in which each cipher that is broken reveals the clue needed to unlock the next stage. The objective is to gain practical experience with classical cryptographic techniques and to understand why they are considered insecure by modern standards.

The work covers three classical ciphers of increasing complexity: the Caesar cipher, where a single shift is applied to every letter; the substitution cipher, where each letter is replaced by a fixed different letter; and the Vigenère cipher, where a repeating keyword determines a different shift for each position.

For each cipher the lab asks both to use the cipher (encrypt and decrypt with a known key) and to attack it (recover the plaintext without knowing the key). The attacks progress from brute force, which works when the key space is small, to frequency analysis, which exploits the statistical structure of English.

The report is organised in three parts corresponding to the three stages of the escape room: the warm-up (Caesar), Room 1 (substitution) and Room 2 (Vigenère).

Part 1 – Warm-Up: Caesar Cipher

0.1 What the part is about

The warm-up introduces the Caesar cipher, the simplest classical cipher. A single key K shifts every letter of the alphabet by K positions. The goal is to encrypt and decrypt messages with a known key, and then to write a brute-force program that recovers the plaintext when the key is unknown.

0.2 What we did

- **Exercise 1 – Encrypt with $K = 4$:** Wrote a function that shifts each letter forward by K positions in the alphabet, wrapping around after Z. Applied it to the plaintext CRYPTO IS FUN UNTIL THE PROFESSOR SAYS QUIZ.
- **Exercise 2 – Decrypt with $K = 8$:** Applied the reverse operation (shift backward by 8) to the ciphertext GWCL JMBBMZ VWB JM CAQVO BWBTA BW AWTDM BPQA.
- **Exercise 3 – Brute force:** Since Caesar has only 26 possible keys, wrote a program that tries every shift from 0 to 25 and prints all candidate plaintexts, then identified the readable one.

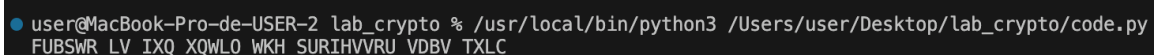
0.3 Difficulties encountered

No major difficulties. The only point of attention was handling the wrap-around correctly when a shifted letter goes past Z, which is solved by the modulo operation ($\% 26$).

0.4 Results / Evidence

Exercise 1 produced the ciphertext:

GVCTXS MW JYR YRXMP XLI TVSJIWWSV WECW UYMD



```
user@MacBook-Pro-de-USER-2 lab_crypto % /usr/local/bin/python3 /Users/user/Desktop/lab_crypto/code.py
FUBSWR LV IXQ XQWLO WKH SURIHVVRU VDBV TXLC
```

Figure 1: Caesar encryption ($K = 4$) and decryption ($K = 8$) output.

Exercise 2 recovered the plaintext:

YOU'D BETTER NOT BE USING TOOLS TO SOLVE THIS

Exercise 3: The brute-force program printed all 26 candidate plaintexts. At shift 11 the output is readable English.

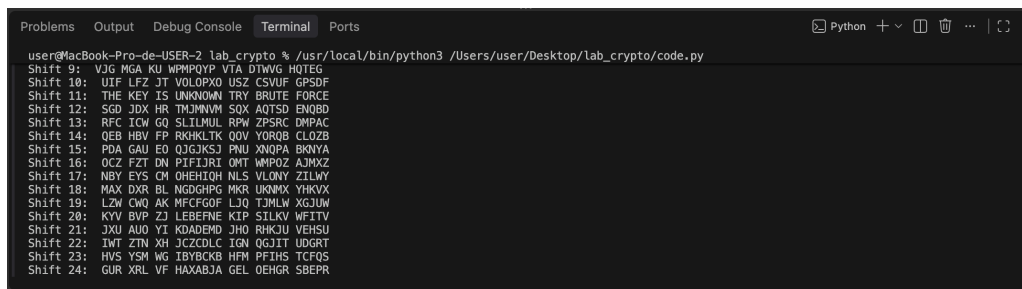


Figure 2: Brute-force output: shift 11 gives THE KEY IS UNKNOWN TRY BRUTE FORCE.

Part 2 – Room 1: Substitution Cipher

0.5 What the part is about

Room 1 introduces the monoalphabetic substitution cipher, where each letter of the alphabet is replaced by a different fixed letter. The part asks to encrypt and decrypt with a known substitution alphabet, to understand why brute force is impractical (the key space is 26!), and to break an unknown substitution using frequency analysis.

0.6 What we did

- **Exercise 1 – Encrypt:** Using the given substitution alphabet, encrypted KEEP THE SECRET SAFE by looking up each plaintext letter and replacing it with its cipher equivalent.
- **Exercise 2 – Decrypt:** Reversed the process on ERVYE DXVR SXFPJ by looking up each ciphertext letter in the cipher alphabet and returning the corresponding plain letter.
- **Exercise 3 – Key space:** Calculated $26! \approx 4.03 \times 10^{26}$ and explained that brute force is impractical. Instead, an attacker can compare ciphertext letter frequencies to known English frequencies (E $\approx$ 12.7%, T $\approx$ 9.1%, etc.) to build a partial mapping, then refine it using common words and patterns.
- **Exercise 4 – Break the cipher:** Wrote a frequency-analysis program that counts each letter in the ciphertext, calculates its percentage, and ranks letters from most to least frequent. Applied the program to the intercepted message and built the full mapping in two stages: first by aligning the frequency ranking with the English hint, then by identifying common words and patterns in the partially decrypted text.

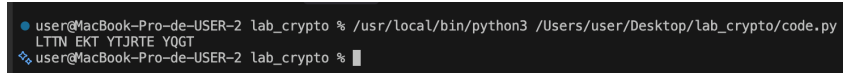
0.7 Difficulties encountered

Frequency analysis alone did not give the complete key. After mapping the most frequent ciphertext letters to the most frequent English letters, several letters remained ambiguous. These were resolved by examining word patterns: for example, recognising EKT as THE, BTYYQFT as MESSAGE, and NQRQFRQNK as PARAGRAPH provided the remaining letter mappings.

0.8 Results / Evidence

Exercise 1 produced:

LTTN EKT YTJRTE YQGT



```
user@MacBook-Pro-de-USER-2 lab_crypto % /usr/local/bin/python3 /Users/user/Desktop/lab_crypto/code.py
LTTN EKT YTJRTE YQGT
user@MacBook-Pro-de-USER-2 lab_crypto %
```

Figure 3: Substitution encryption output.

Exercise 2 recovered:

TRUST YOUR LOGIC

Exercise 3 – Frequency mapping used:

Cipher	T	E	R	Y	X	Q	K	O	Z	P	F
Plain	E	T	R	S	O	A	H	N	D	I	G

The remaining mappings were completed by pattern matching: $B \rightarrow M$, $N \rightarrow P$, $V \rightarrow U$, $D \rightarrow Y$, $H \rightarrow V$, $I \rightarrow W$, $G \rightarrow F$, $L \rightarrow K$, $J \rightarrow C$, $C \rightarrow Q$, $M \rightarrow B$.

Exercise 4 – Decrypted message:

THE SECRET MESSAGE IS HIDDEN IN THIS PARAGRAPH. LETTER FREQUENCY DOES NOT REVEAL THE WHOLE KEY, BUT IT GIVES A GOOD START. LOOK FOR COMMON WORDS, REPEATED PATTERNS, AND SHORT WORDS. WHEN YOU RECOVER THE MESSAGE, THE VIGENERE KEY HAS FOUR LETTERS.

Clue for Room 2: *The Vigenère key has four letters.*

Part 3 – Room 2: Vigenère Cipher

0.9 What the part is about

Room 2 introduces the Vigenère cipher. Unlike Caesar, which uses a single shift, Vigenère uses a keyword that repeats over the plaintext: each letter of the key determines the Caesar shift for the corresponding position. The goal is to recover the key and decrypt the intercepted ciphertext, using the clue from Room 1 (the key has four letters).

0.10 What we did

- **Step 1 – Split into groups:** Since the key has 4 letters, every 4th character is encrypted by the same key letter. Split the ciphertext into four groups:

```
group1 = ciphertext[0::4]
group2 = ciphertext[1::4]
group3 = ciphertext[2::4]
group4 = ciphertext[3::4]
```

- **Step 2 – Frequency analysis per group:** Reused the frequency-analysis program on each group separately. Since all letters in one group are shifted by the same amount, each group is effectively a Caesar cipher. Assumed the most frequent letter in each group corresponds to plaintext E, and calculated the shift.
- **Step 3 – Recover key:** Converted each shift into a key letter:

Group	Most frequent letter	Shift	Key letter
1	G	2	C
2	S	14	O
3	H	3	D
4	I	4	E

- **Step 4 – Decrypt:** Used the recovered key CODE to shift each ciphertext letter backward by the corresponding key letter's value.

0.11 Difficulties encountered

The main difficulty was understanding how splitting the ciphertext reduces Vigenère to multiple Caesar ciphers. Once this relationship was clear, the same frequency-analysis tool from Room 1 could be reused directly on each group. No other issues were encountered.

0.12 Results / Evidence

Recovered key: CODE

Decrypted message:

THE SECRET KEEPER LEFT THESE NOTES BECAUSE THE LETTER
E APPEARS EVERYWHERE. EVERY TEAM MEMBER SHOULD CHECK
EACH COLUMN. SEEK THE MOST FREQUENT SYMBOL AND USE IT
TO RECOVER THE KEY. WHEN THE MESSAGE BECOMES CLEAR,
ENTER THE ACCESS WORD: ECLIPSE.

```

[10:35]
THE SECRET MESSAGE IS HIDDEN IN THIS PARAGRAPH. LETTER FREQUENCY DOES NOT REVEAL THE WHOLE KEY, BUT IT GIVES A GOOD START. LOOK FOR COMMON WORDS, REPEA
TED PATTERNS, AND SHORT WORDS. WHEN YOU RECOVER THE MESSAGE, THE VIGENERE KEY HAS FOUR LETTERS.
Most frequent letters: G S H I
Key: CODE
Plaintext: THESECRETKEEPERLEFTTHESENOTESBECAUSETHELETTEREAPPEARSEVERYWHEREEVERYTEAMMEMBERSHOULDCHECKEACHCOLUMNSEEKTHEMOSTFREQUENTSMBOLANDUSEITTORECOVE
RTHEKEYWHENTHEMESSAGEBECOMESCLEARENTERTHEACCESSWORDECLIPSE
user@MacBook-Pro-de-USER-2 lab_crypto %
```

Figure 4: Vigenère attack output: recovered key **CODE** and decrypted plaintext ending with the access word **ECLIPSE**.

Access word / Flag: ECLIPSE

Conclusion

What we learned about classical ciphers

This lab showed that all three classical ciphers are insecure, each for a clear reason:

The Caesar cipher has only 26 possible keys, so a simple brute-force loop recovers the plaintext instantly. The substitution cipher has a much larger key space ($26! \approx 4 \times 10^{26}$), which makes brute force impractical, but it preserves the frequency distribution of the plaintext language. An attacker can compare letter frequencies in the ciphertext to known English frequencies and recover most of the key. The Vigenère cipher hides the frequency pattern by using multiple shifts, but once the key length is known, the ciphertext splits into independent Caesar ciphers that are each vulnerable to the same frequency analysis.

The common weakness across all three is that they rely on simple, reversible transformations of individual letters. None of them diffuse information across the message or make the relationship between key and ciphertext complex enough to resist statistical analysis.

How this work can be improved

On the attack side, the frequency analysis could be made more robust by also analysing bigrams (two-letter pairs like TH, HE, IN) and trigrams (THE, AND, ING), which would speed up the substitution attack and reduce the manual pattern-matching step. For Vigenère, the Kasiski examination or the index of coincidence could be used to determine the key length automatically when it is not given as a clue.

On the defence side, the lab illustrates why modern cryptography moved away from letter-by-letter substitution toward block ciphers (AES) and public-key systems (RSA), which operate on large blocks of bits and rely on computational hardness rather than key secrecy alone.